MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, ANALOG, KEY CODE NQD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.57972 failures per million operating hours for class I or 2.2629 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NQD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/325-1 for class I and M28787/325-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

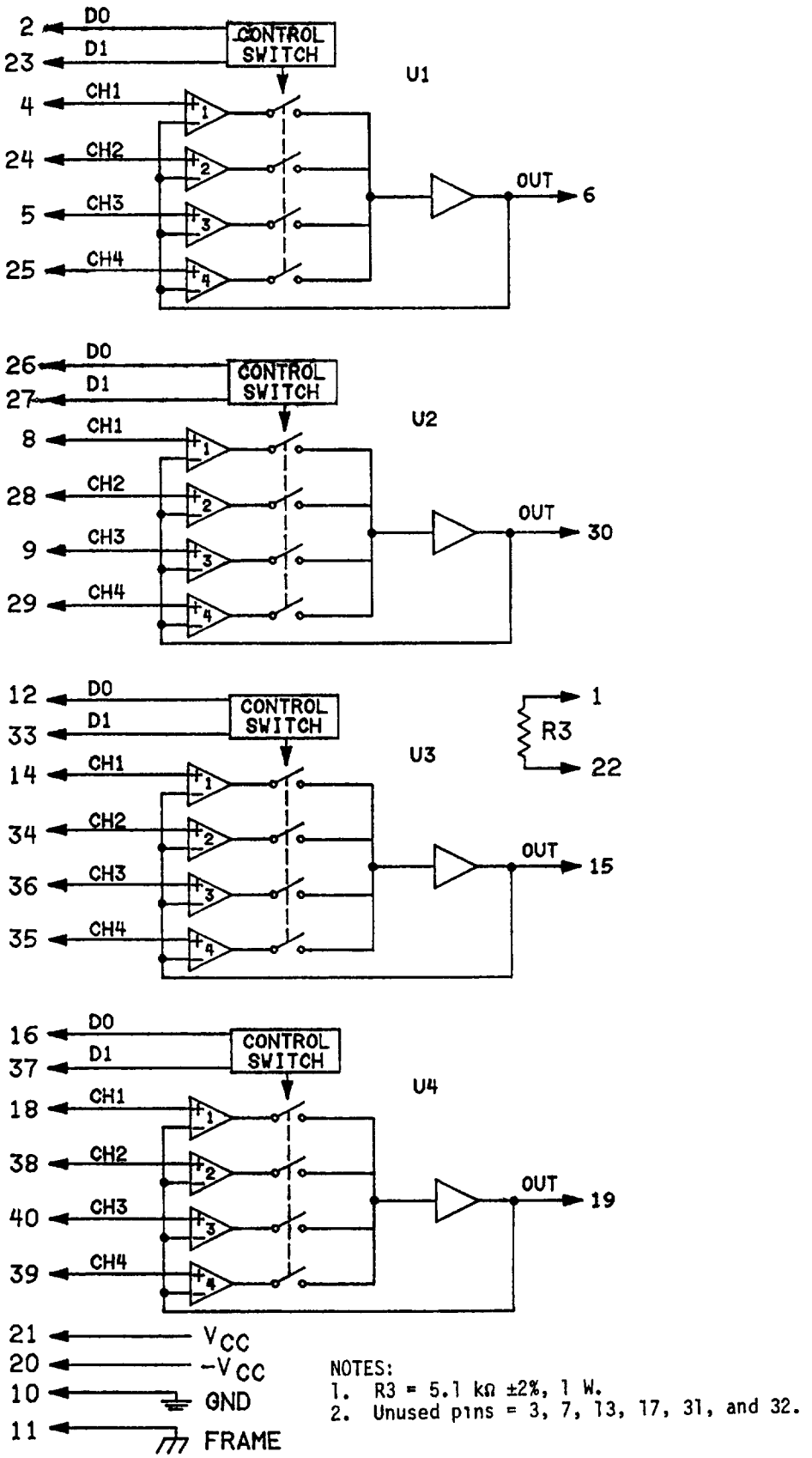
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. R3 = 5.1 kΩ ±2%, 1 W.
2. Unused pins = 3, 7, 13, 17, 31, and 32.

FIGURE 1. NQD module function diagram, multiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	V_s stress	III	Q	← Must pass remaining tests →										
B1	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B2-B7	Isolation													
	Initial	IV	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C4	Output short circuit current	V	S	5	40	5	40	5	40	5	40	5	40	mA
D1,D2	Power supply current	VI	S	1	32	1	32	1	32	1	32	1	32	mA
E1-E4	Power supply rejection ratio check	VII	S	-∞	-52	-∞	-52	-∞	-52	-∞	-54	-∞	-52	dB
F1	Resistance													
	Initial	VIII	X	4.99	5.21	4.97	5.23	4.96	5.24	4.91	5.29	4.93	5.27	kΩ
	End-of-life	VIII	X	4.89	5.31	4.87	5.33	4.86	5.34	4.81	5.39	4.83	5.37	kΩ
F2	Resistance	VIII	X	0	1	0	1	0	1	0	1	0	1	Ω
G1-G8	Input current, high level	IX	X	0	15	0	15	0	15	0	15	0	15	nA
H1-H8	Input current, low level	IX	X	0.1	1.6	0.1	1.6	0.1	1.6	0.1	1.6	0.1	1.6	mA
J1-J16	Input offset voltage	X	X	-7	7	-7	7	-7	7	-7	7	-7	7	mV
K1-K16	Gain (100 kHz) check	XI	S	-3	3	-3	3	-3	3	-3	3	-3	3	dB
L1-L32	Output slew rate check	XII	X	0.9	80.0	0.9	80.0	0.9	80.0	0.9	80.0	0.9	80.0	V/μs
M1-M16	Distortion	XIII	X	0	1	0	1	0	1	0	1	0	1	%
N1-N16	Crosstalk check	XIV	Q	-∞	-50	-∞	-50	-∞	-50	-∞	-50	-∞	-50	dB

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P16	Functional	XV	X	2.35	2.45	2.35	2.45	2.35	2.45	2.35	2.45	2.35	2.45	V
Q1,Q2	Power supply filter capacitance	XVI	X	0.75	1.25	0.75	1.25	0.75	1.25	0.70	1.30	0.70	1.30	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through F and H through Q shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test condition G shall be performed at 25°C, 60°C, and 85°C using the initial limits for the applicable temperature.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B and D through Q shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	1.2	1.2	W
Supply voltages (see 6.5.1)	±20	±20	V

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	Class I	Class II	Unit
Output voltage swing	±10	±10	V
Input levels:			
Digital threshold levels			
Low level	0.5	0.5	V
High level	2.4	2.4	V
Analog (maximum)	±20	±20	V
Minimum Bandwidth	100	100	kHz

6.5 Module function. This module contains four programmable amplifiers. For each amplifier two digital control lines are used to select one of four unity gain buffer amplifiers. The selected buffer is connected to a common output as shown on figure 1.

6.5.1 Maximum ratings. The maximum ratings for supply voltages and input levels given in 6.4 represent extremes beyond which damage to the module may occur. Operation at the extremes is performed in this specification to ensure that the module will survive these levels without damage.

6.5.2 Module handling procedures. This module contains devices which are susceptible to damage when exposed to a static electrical charge. When handling, shipping, or storing, all module pins should be shorted or the module should be placed in an electrically conductive package. Soldering iron tips and all test equipment must be grounded. Failure to comply with these guidelines could cause irreversible module damage.

6.5.3 Input to output relationship. The digital input versus ON channel relationship is shown in table II.

TABLE II. Input to output relationship. 1/ 2/

D1	D0	Selected channel
L	L	1
L	H	2
H	L	3
H	H	4

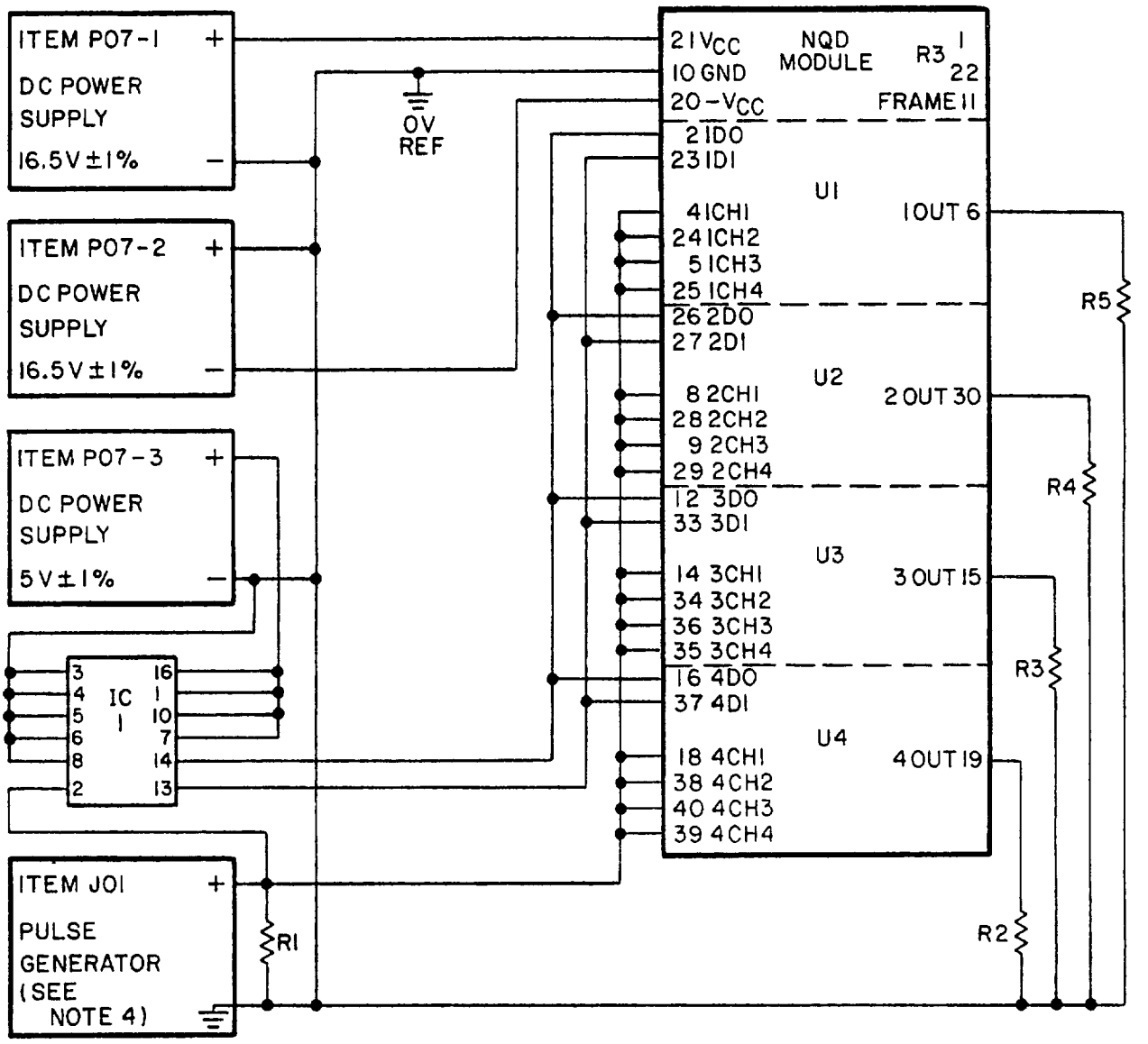
1/ L = Low level = 0.5 V max.
2/ H = High level = 2.4 V min.

Agent:
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Preparing activity:
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2-R5 = 2 kΩ ±5%, 1/4 W.
3. IC1 = 74LS161 or equivalent
4. Item J01 Setup:
   PA = 4.5 V ±5%.
   Frequency = 100 kHz ±10%.
   Duty cycle = 50 ±2%
   $t_{TLH} = t_{THL} \leq 30$ ns.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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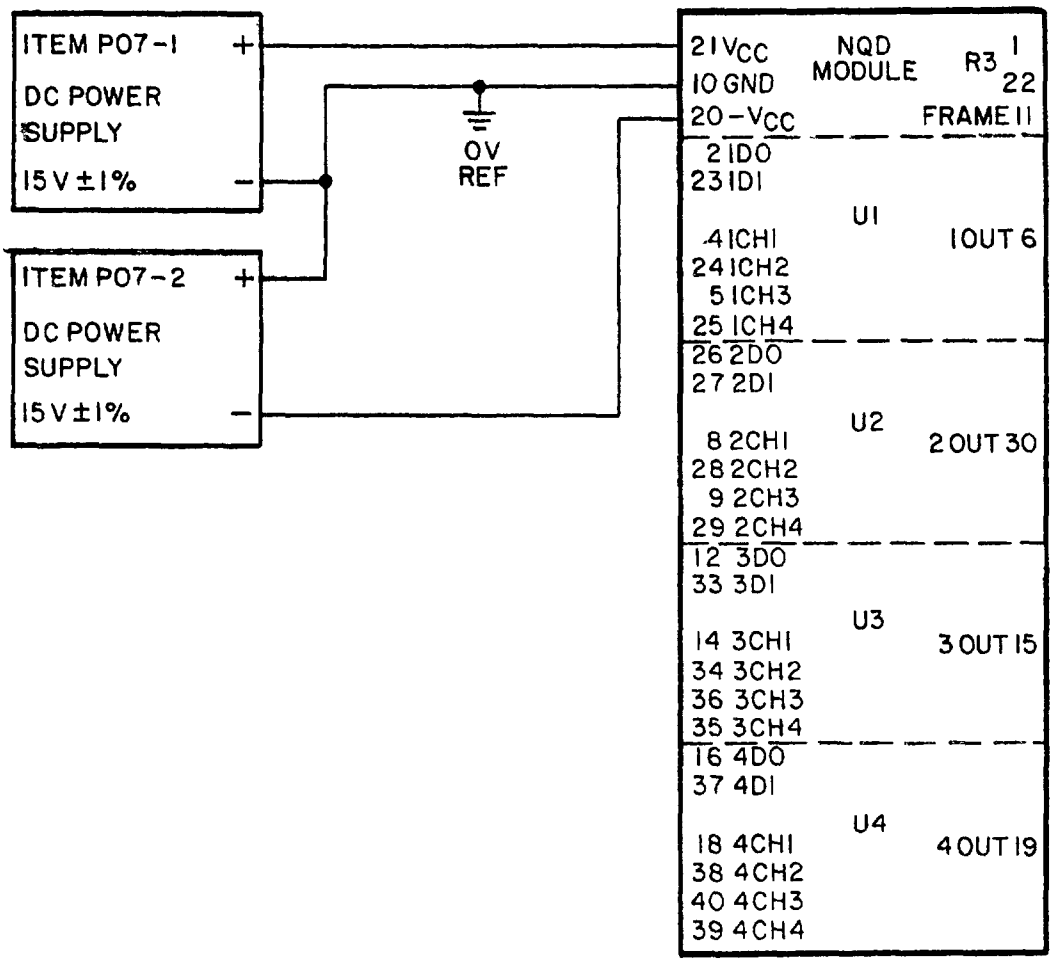


FIGURE 3. Vibration test circuit.

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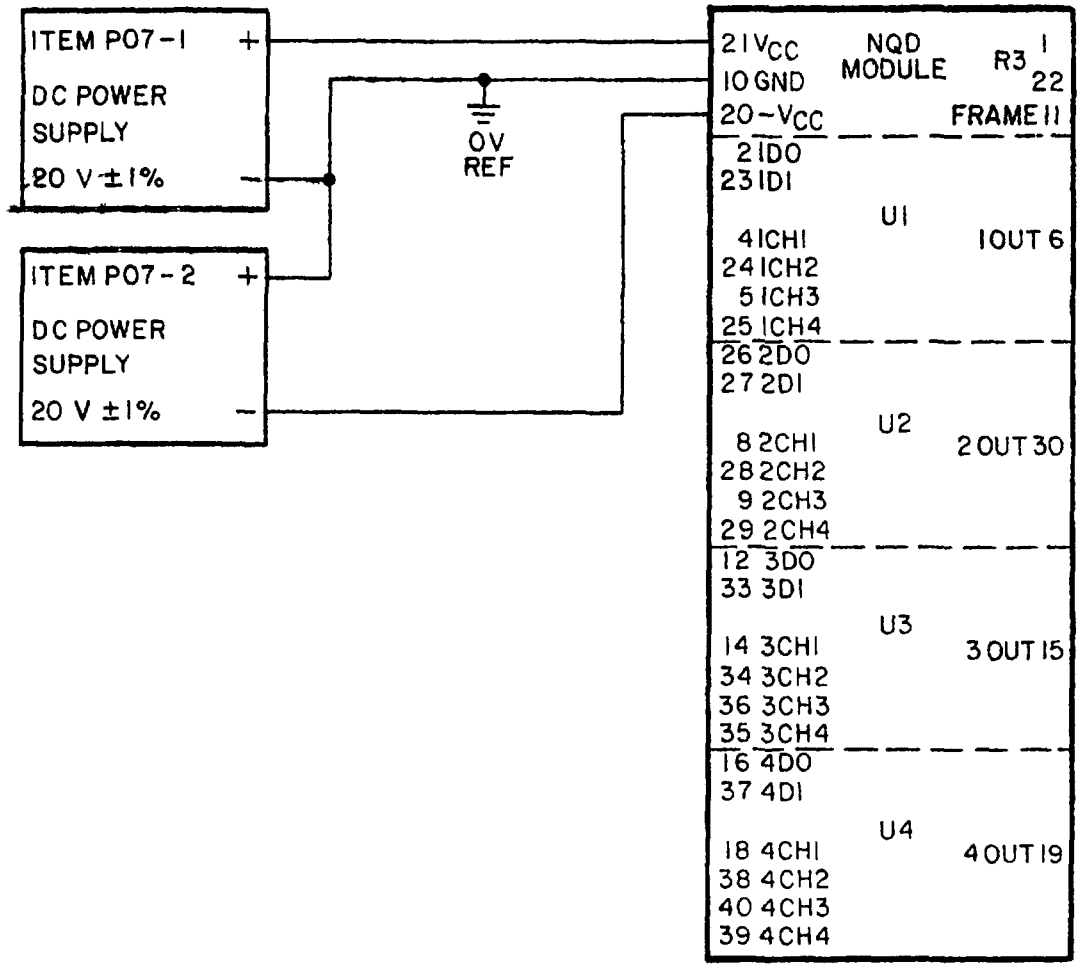
TABLE III. V_S stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
A	V_S stress	Must pass remaining tests			4

TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, frame	To all other pins	11	10	∞	MΩ	5
B2	Isolation, used to unused pins	"	32	"	"	"	"
B3	"	"	3	"	"	"	"
B4	"	"	7	"	"	"	"
B5	"	"	13	"	"	"	"
B6	"	"	17	"	"	"	"
B7	"	"	31	"	"	"	"

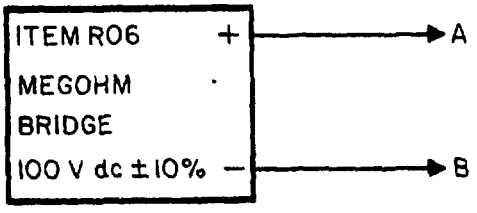
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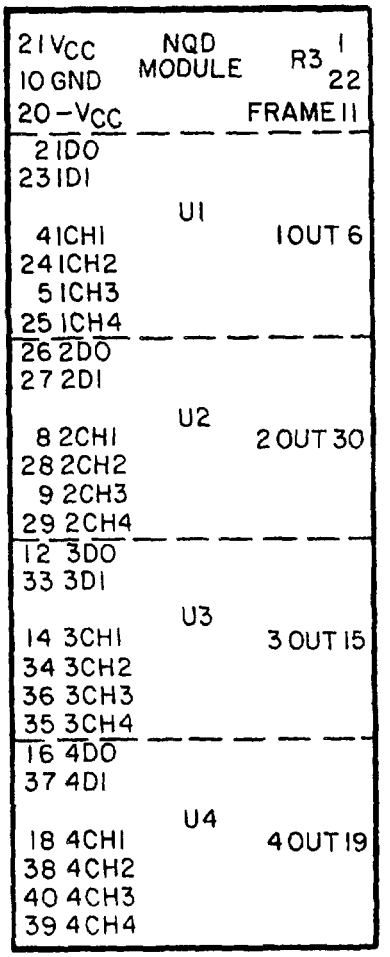


NOTE: Apply power for 1 minute minimum.

FIGURE 4. V_S stress test circuit for table III.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table IV.

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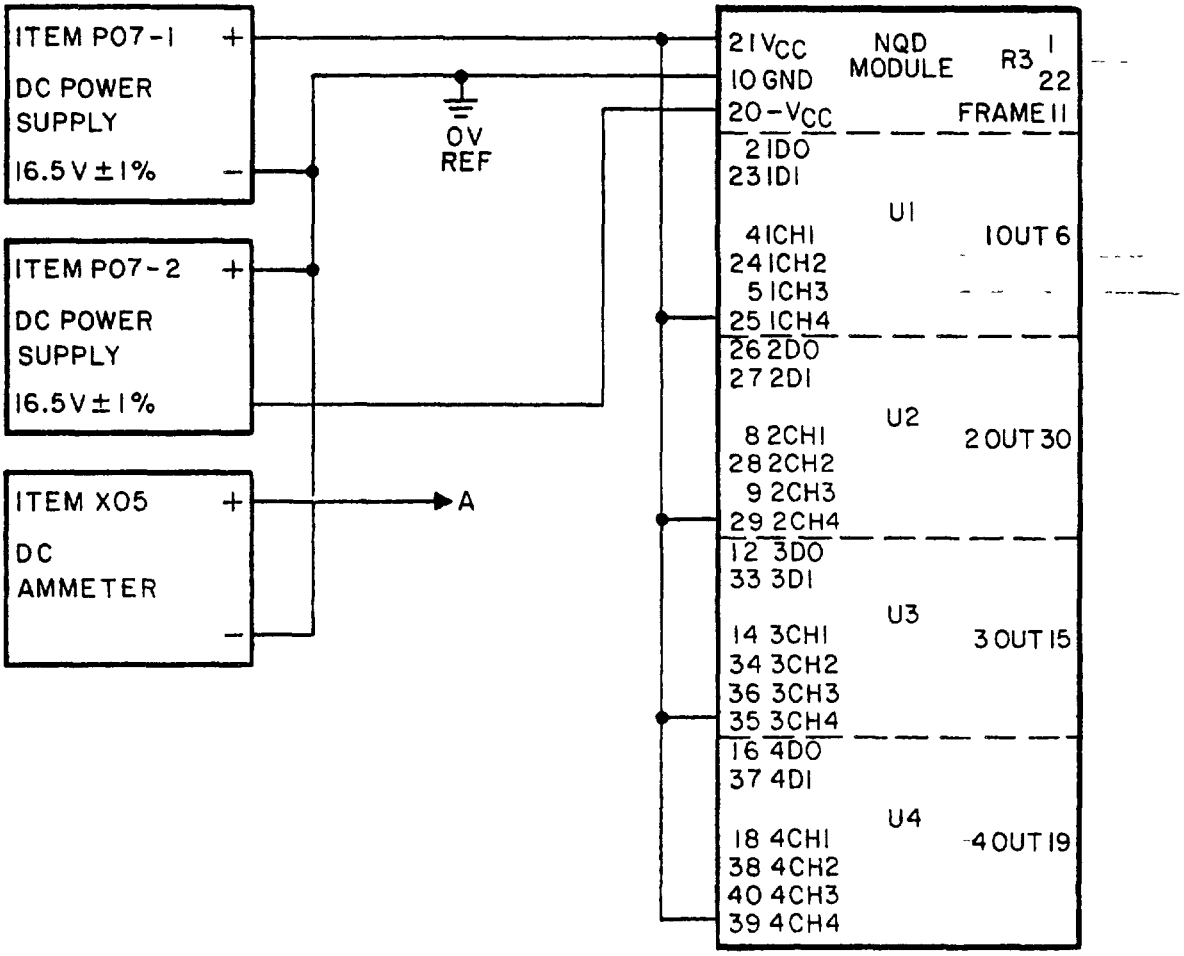
TABLE V. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	Output short circuit current	6	5	40	mA	6
C2	"	30	"	"	"	"
C3	"	15	"	"	"	"
C4	"	19	"	"	"	"

TABLE VI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current	1	32	mA	7
D2	Power supply current	1	32	mA	7

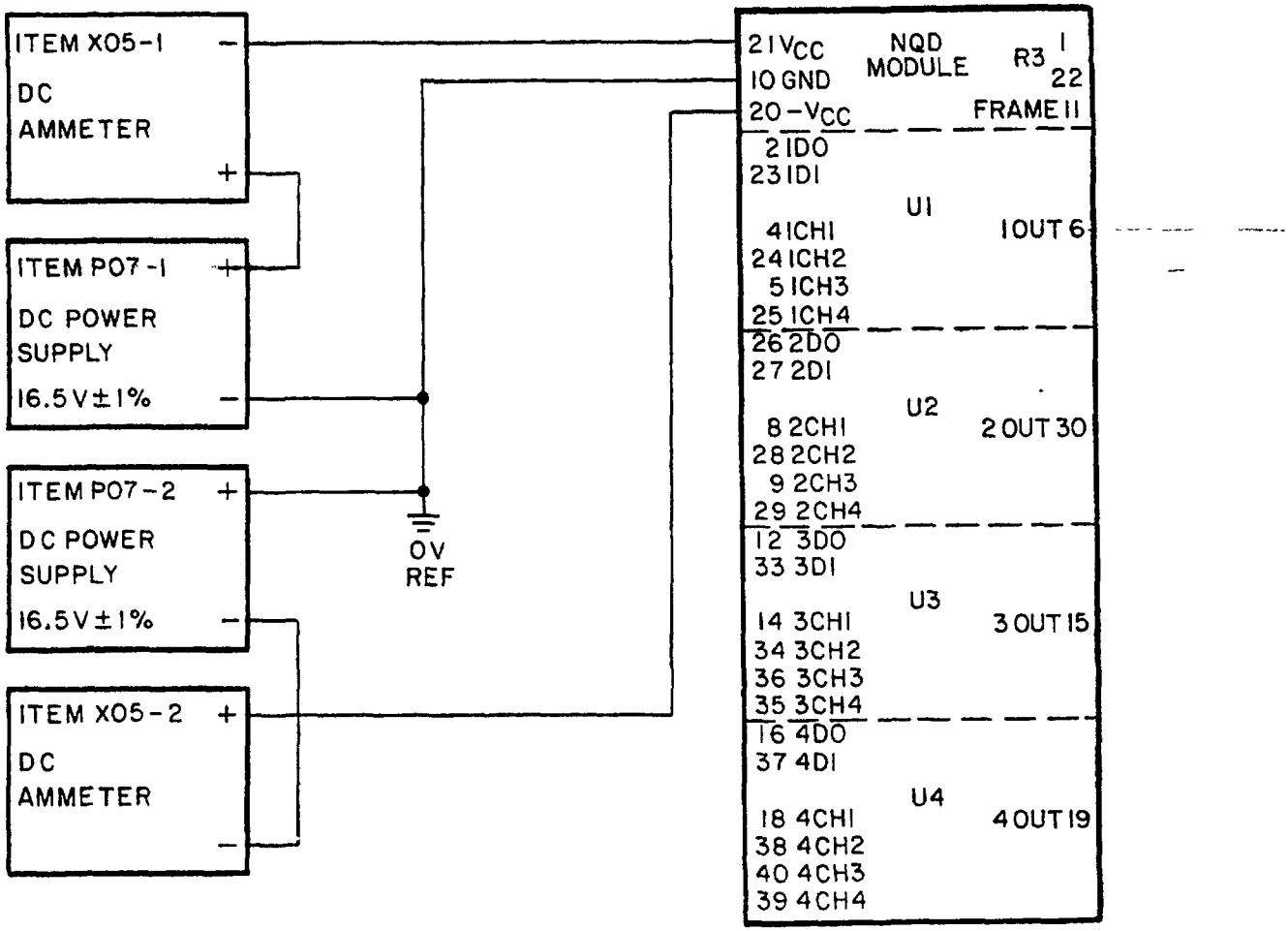
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NOTE: The power shall be applied for a minimum of 1 minute before taking the readings.

FIGURE 6. Output short circuit current test circuit for table V.

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NOTES:
1. Since items X05-1 and X05-2 are not used simultaneously, only one item X05 is required.
2. Read item X05-1 for test condition D1 and item X05-2 for test condition D2.

FIGURE 7. Power supply current test circuit for table VI.

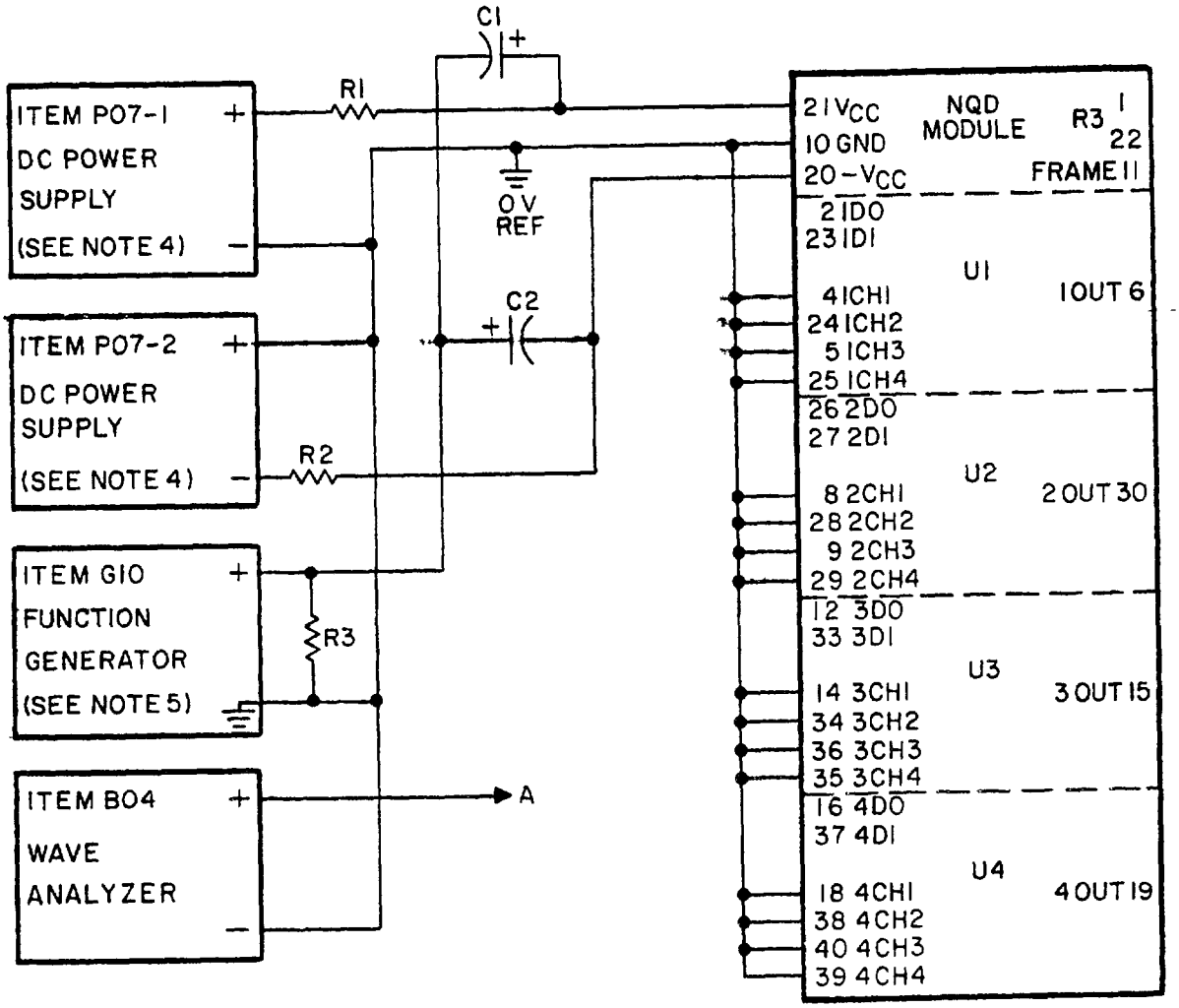
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TABLE VII. Power supply rejection ratio check test. 1/

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Power supply rejection ratio check	6	0.0	2.5	mV rms	8
E2	"	30	"	"	"	"
E3	"	15	"	"	"	"
E4	"	19	"	"	"	"

1/ PSRR (table I, dB) = $20 \log \left(\frac{V_{OUT}}{V_{IN}} \right) = 20 \log \left(\frac{\text{Item B04 reading}}{1 \text{ V rms}} \right)$.

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NOTES:
1. R1 = R2 = 100Ω ±5%, 1/4 W.
2. R3 = 51Ω ±5%, 1 W.
3. C1 = C2 = 20 µF ±20%, 20 V dc, electrolytic.
4. Adjust item P07-1 to obtain 15 V ±1% at pin 21; adjust item P07-2 to obtain 15 V ±1% at pin 20.
5. Adjust item G10 for 1 V rms ±1%, 1 kHz ±1% sine wave at pins 20 and 21.
6. Set item B04 for a 10 Hz bandwidth. Tune to 1 kHz and fine tune for a maximum deflection.

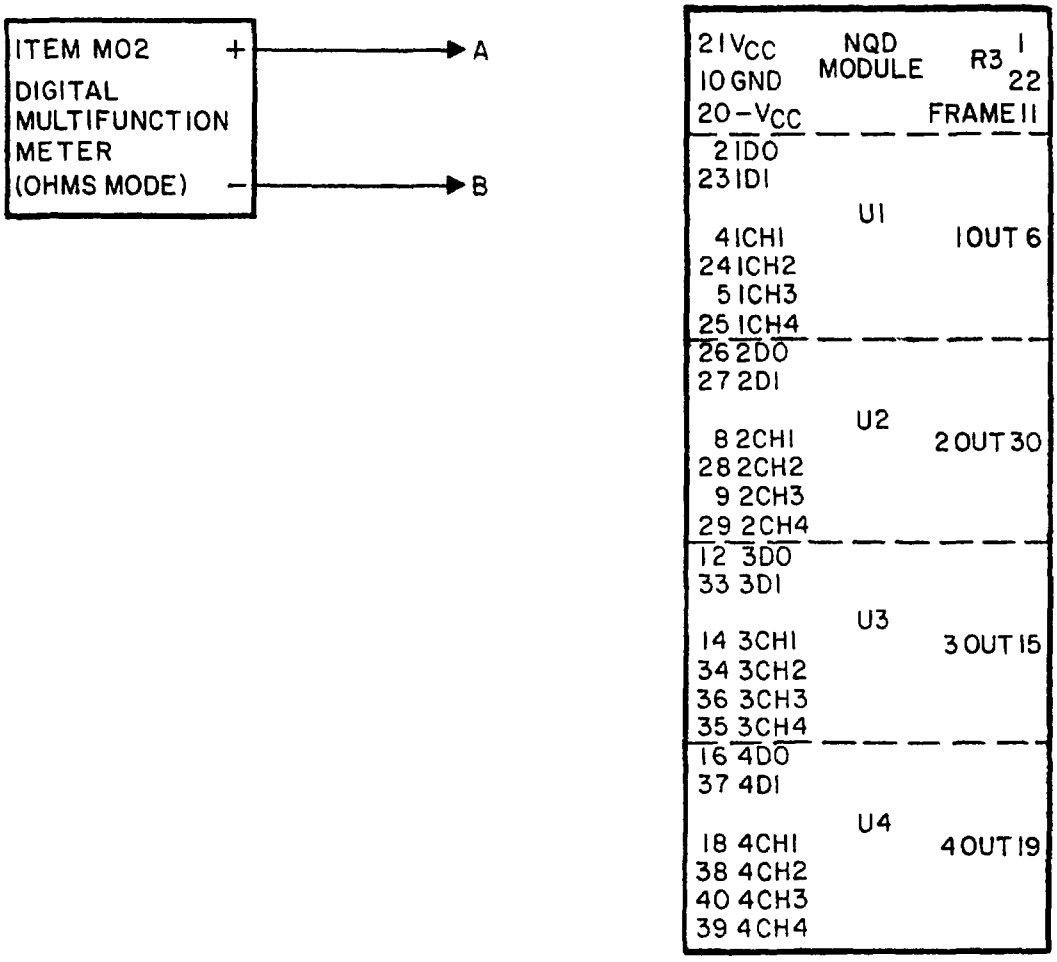
FIGURE 8. Power supply rejection ratio check test circuit for table VII.

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TABLE VIII. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Resistance	1	22	4.99	5.21	kΩ	9
F2	Resistance	11	Frame	0	1	Ω	9

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NOTE: For test condition F2, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

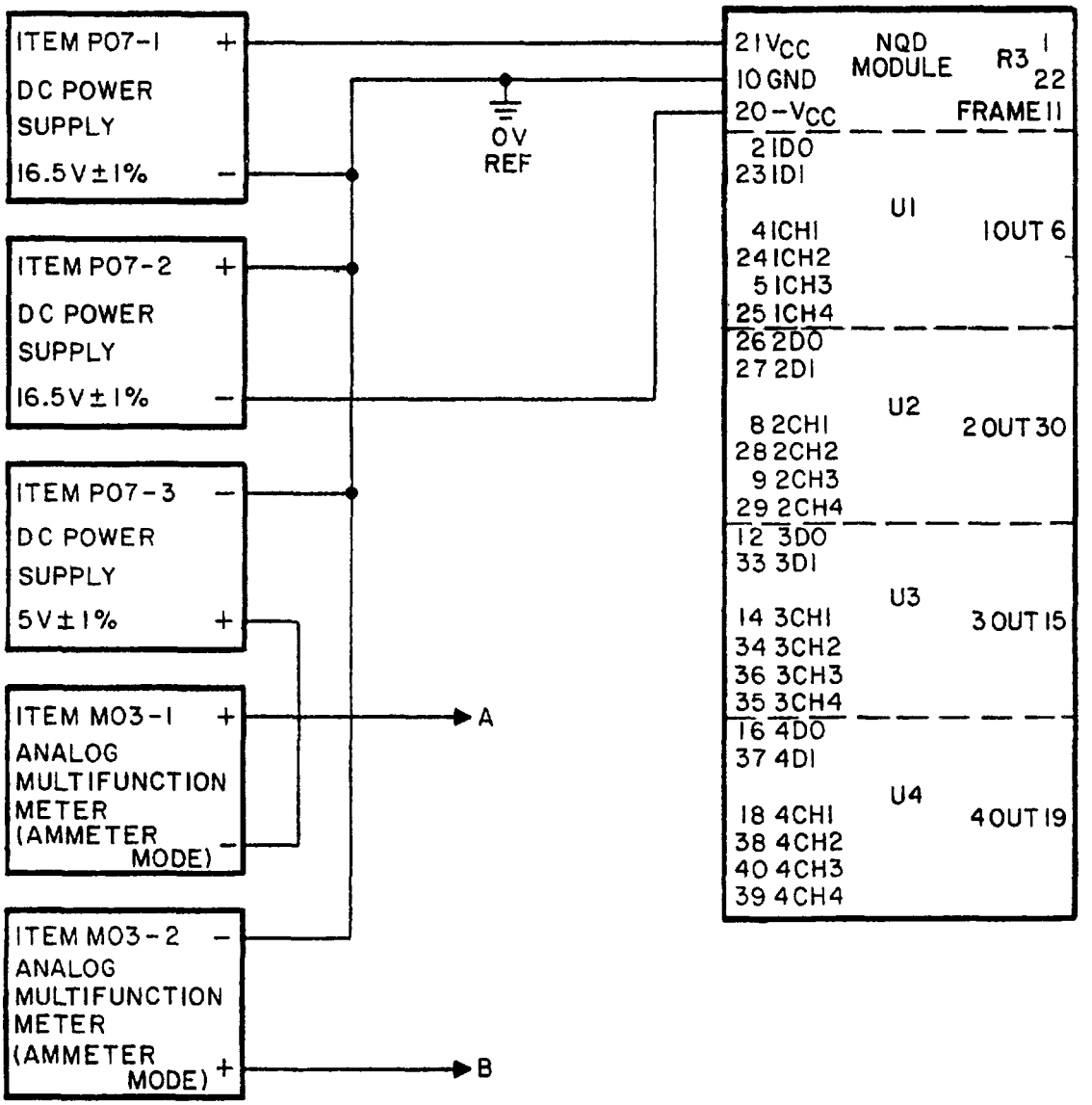
FIGURE 9. Resistance test circuit for table VIII.

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TABLE IX. Input current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	1D0 Input current, high level	2	Open	0	15	nA	10
G2	1D1 "	23	"	"	"	"	"
G3	2D0 "	26	"	"	"	"	"
G4	2D1 "	27	"	"	"	"	"
G5	3D0 "	12	"	"	"	"	"
G6	3D1 "	33	"	"	"	"	"
G7	4D0 "	16	"	"	"	"	"
G8	4D1 "	37	"	"	"	"	"
H1	1D0 Input current, low level	Open	2	0.1	1.6	mA	"
H2	1D1 "	"	23	"	"	"	"
H3	2D0 "	"	26	"	"	"	"
H4	2D1 "	"	27	"	"	"	"
H5	3D0 "	"	12	"	"	"	"
H6	3D1 "	"	33	"	"	"	"
H7	4D0 "	"	16	"	"	"	"
H8	4D1 "	"	37	"	"	"	"

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NOTE: Since items M03-1 and M03-2 are not used simultaneously, only one item M03 is required.

FIGURE 10. Input current (high-level and low-level) test circuit for table IX.

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TABLE X. Input offset voltage test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Input offset voltage	Open	2,23	6	4	-7	7	mV	11
J2	"	2	23	"	"	"	"	"	"
J3	"	23	2	"	"	"	"	"	"
J4	"	2,23	Open	"	"	"	"	"	"
J5	"	Open	27,26	30	8	"	"	"	"
J6	"	26	27	"	"	"	"	"	"
J7	"	27	26	"	"	"	"	"	"
J8	"	27,26	Open	"	"	"	"	"	"
J9	"	Open	12,33	15	14	"	"	"	"
J10	"	12	33	"	"	"	"	"	"
J11	"	33	12	"	"	"	"	"	"
J12	"	33,12	Open	"	"	"	"	"	"
J13	"	Open	16,37	19	18	"	"	"	"
J14	"	16	37	"	"	"	"	"	"
J15	"	37	16	"	"	"	"	"	"
J16	"	16,37	Open	"	"	"	"	"	"

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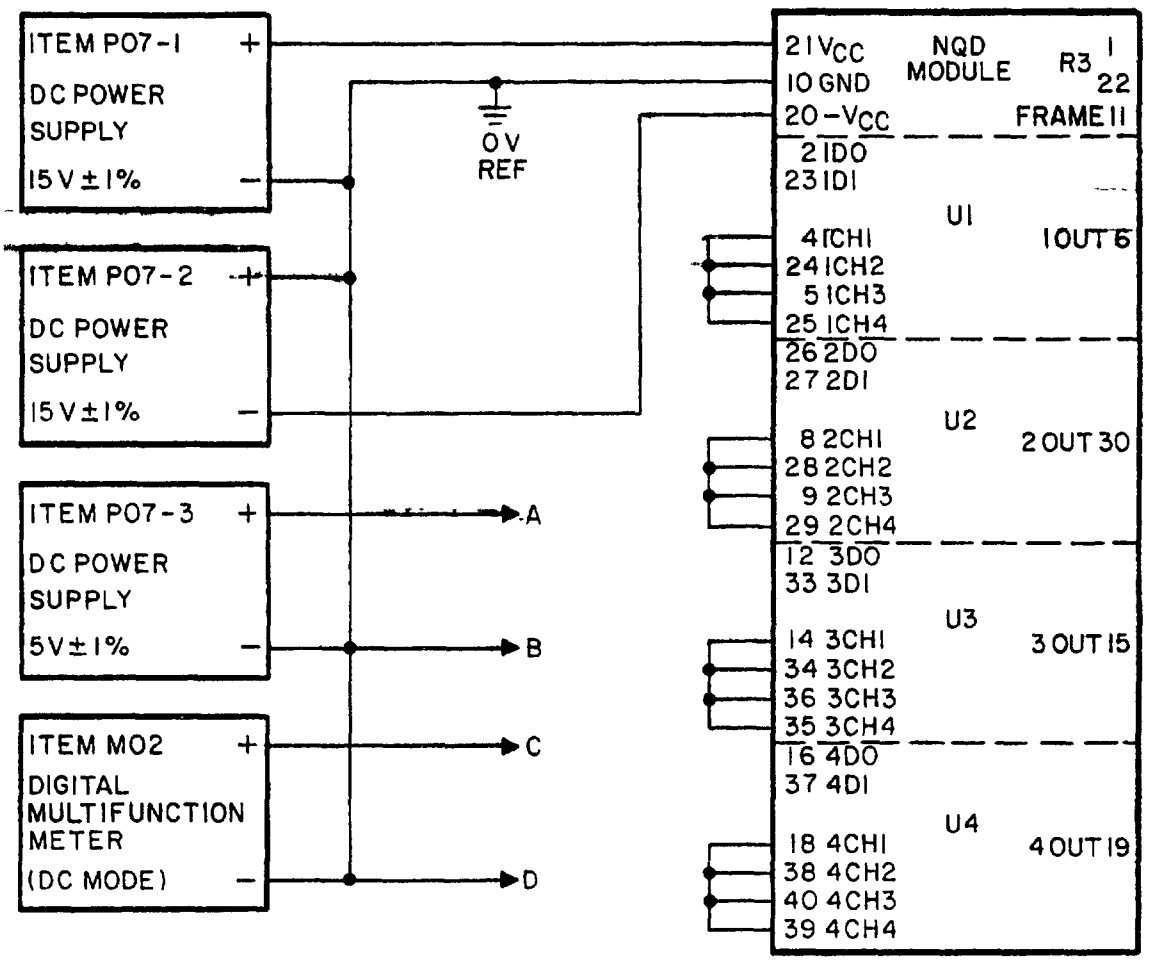


FIGURE 11. Input offset voltage test circuit for table X.

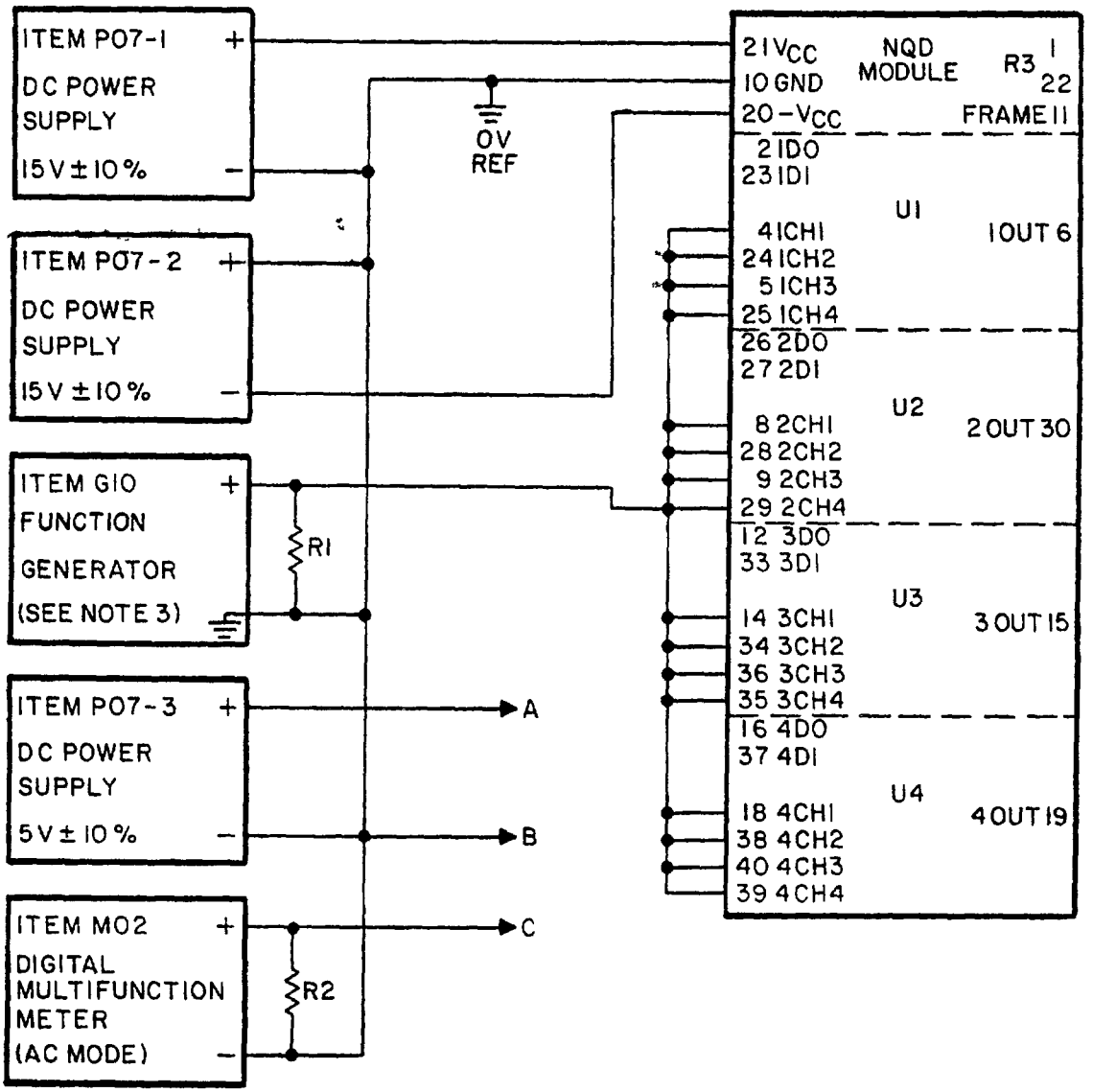
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TABLE XI. Gain (100 kHz) check test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Gain (100 kHz) check	Open	2,23	6	0.0707	0.1414	V rms	12
K2	"	2	23	"	"	"	"	"
K3	"	23	2	"	"	"	"	"
K4	"	2,23	Open	"	"	"	"	"
K5	"	Open	27,26	30	"	"	"	"
K6	"	26	27	"	"	"	"	"
K7	"	27	26	"	"	"	"	"
K8	"	27,26	Open	"	"	"	"	"
K9	"	Open	33,12	15	"	"	"	"
K10	"	12	33	"	"	"	"	"
K11	"	33	12	"	"	"	"	"
K12	"	33,12	Open	"	"	"	"	"
K13	"	Open	16,37	19	"	"	"	"
K14	"	16	37	"	"	"	"	"
K15	"	37	16	"	"	"	"	"
K16	"	16,37	Open	"	"	"	"	"

1/ This test verifies a minimum bandwidth of 100 kHz.
2/ Gain (table I, dB) = $20 \log\left(\frac{\text{Item M02 reading}}{0.1 \text{ V rms}}\right)$.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 2 kΩ ±5%, 1/4 W.
3. Item G10 setup:
   Amplitude = 0.1 V rms ±1%.
   Frequency = 100 kHz ±1%.
   Sine wave.

FIGURE 12. Gain (100 kHz) check test circuit for table XI.

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TABLE XII. Output slew rate check test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Output slew rate check	Open	2,23	6	0.05	4.44	µs	13,14
L2	"	Open	2,23	"	"	"	"	"
L3	"	2	23	"	"	"	"	"
L4	"	2	23	"	"	"	"	"
L5	"	23	2	"	"	"	"	"
L6	"	23	2	"	"	"	"	"
L7	"	2,23	Open	"	"	"	"	"
L8	"	2,23	Open	"	"	"	"	"
L9	"	Open	26,27	30	"	"	"	"
L10	"	Open	26,27	"	"	"	"	"
L11	"	26	27	"	"	"	"	"
L12	"	26	27	"	"	"	"	"
L13	"	27	26	"	"	"	"	"
L14	"	27	26	"	"	"	"	"
L15	"	26,27	Open	"	"	"	"	"
L16	"	26,27	Open	"	"	"	"	"
L17	"	Open	12,33	15	"	"	"	"
L18	"	Open	12,33	"	"	"	"	"
L19	"	12	33	"	"	"	"	"
L20	"	12	33	"	"	"	"	"
L21	"	33	12	"	"	"	"	"
L22	"	33	12	"	"	"	"	"

See footnotes at end of table.

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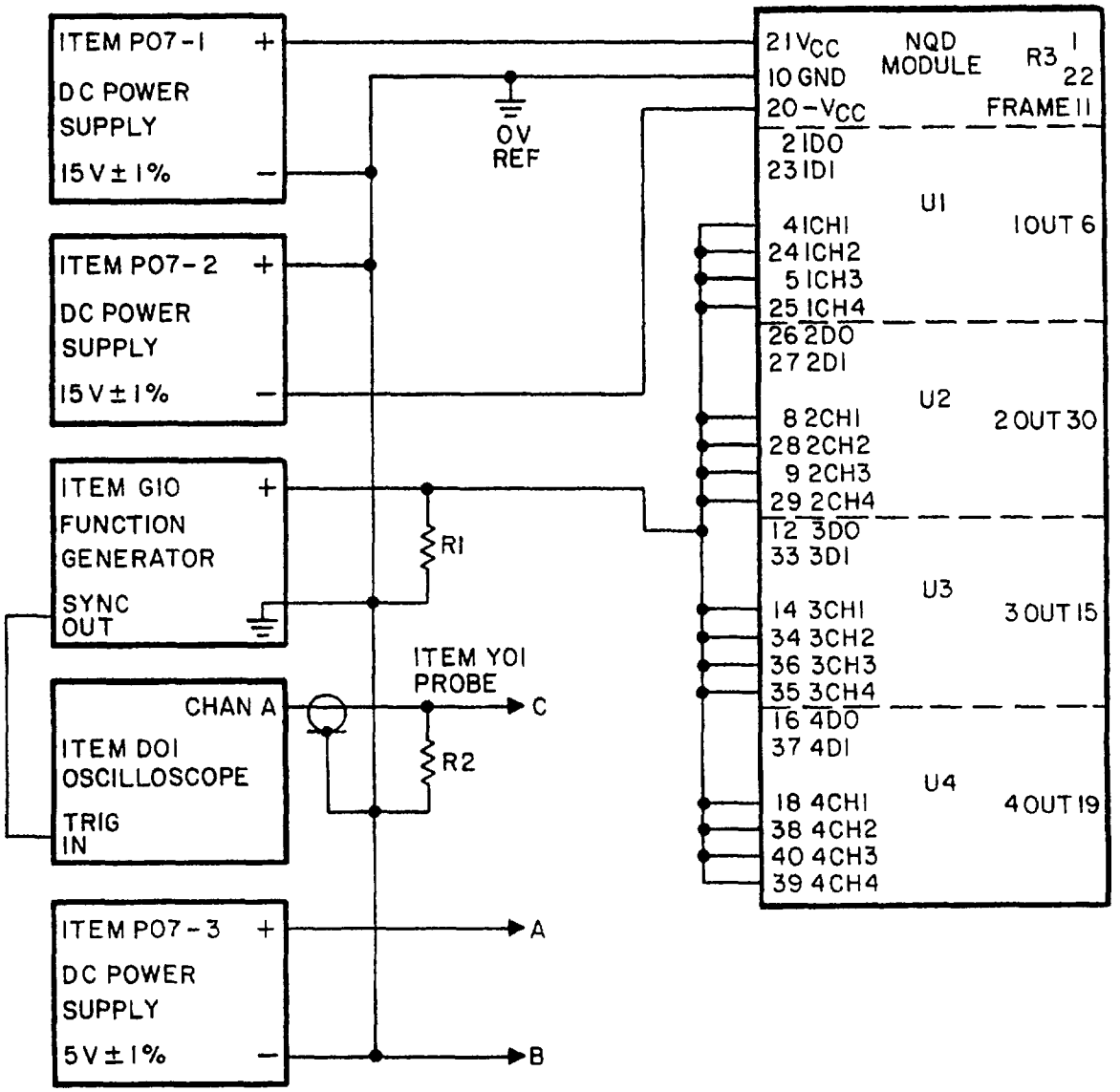
TABLE XII. Output slew rate check test 1/ 2/ - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L23	Output slew rate check	12,33	Open	15	0.05	4.44	µs	13,14
L24	"	12,33	Open	15	"	"	"	"
L25	"	Open	16,37	19	"	"	"	"
L26	"	Open	16,37	"	"	"	"	"
L27	"	16	37	"	"	"	"	"
L28	"	16	37	"	"	"	"	"
L29	"	37	16	"	"	"	"	"
L30	"	37	16	"	"	"	"	"
L31	"	16,37	Open	"	"	"	"	"
L32	"	16,37	Open	"	"	"	"	"

1/ Odd test conditions L1-L31 are negative slew rates.
Even test conditions L2-L32 are positive slew rates.

2/ Slew rate (table I, V/µs) $= \left(\frac{4V}{\text{Test condition L}}\right)$.

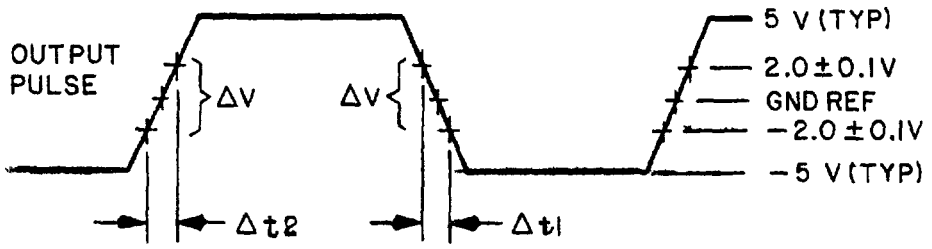
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 2 kΩ ±5%, 1/4 W.
3. Item G10 setup:
   Amplitude = 10 V_{p-p} ±1%.
   Frequency = 1 kHz ±1%.
   Square wave.

FIGURE 13. Output slew rate check test circuit for table XII.

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NOTES:
1. ΔV/Δt = Output slew rate.
2. Δt1 = L1 - L31 (odd), negative slew rate.
3. Δt2 = L2 - L32 (even), positive slew rate.

FIGURE 14. Output slew rate check test circuit waveform for table XII.

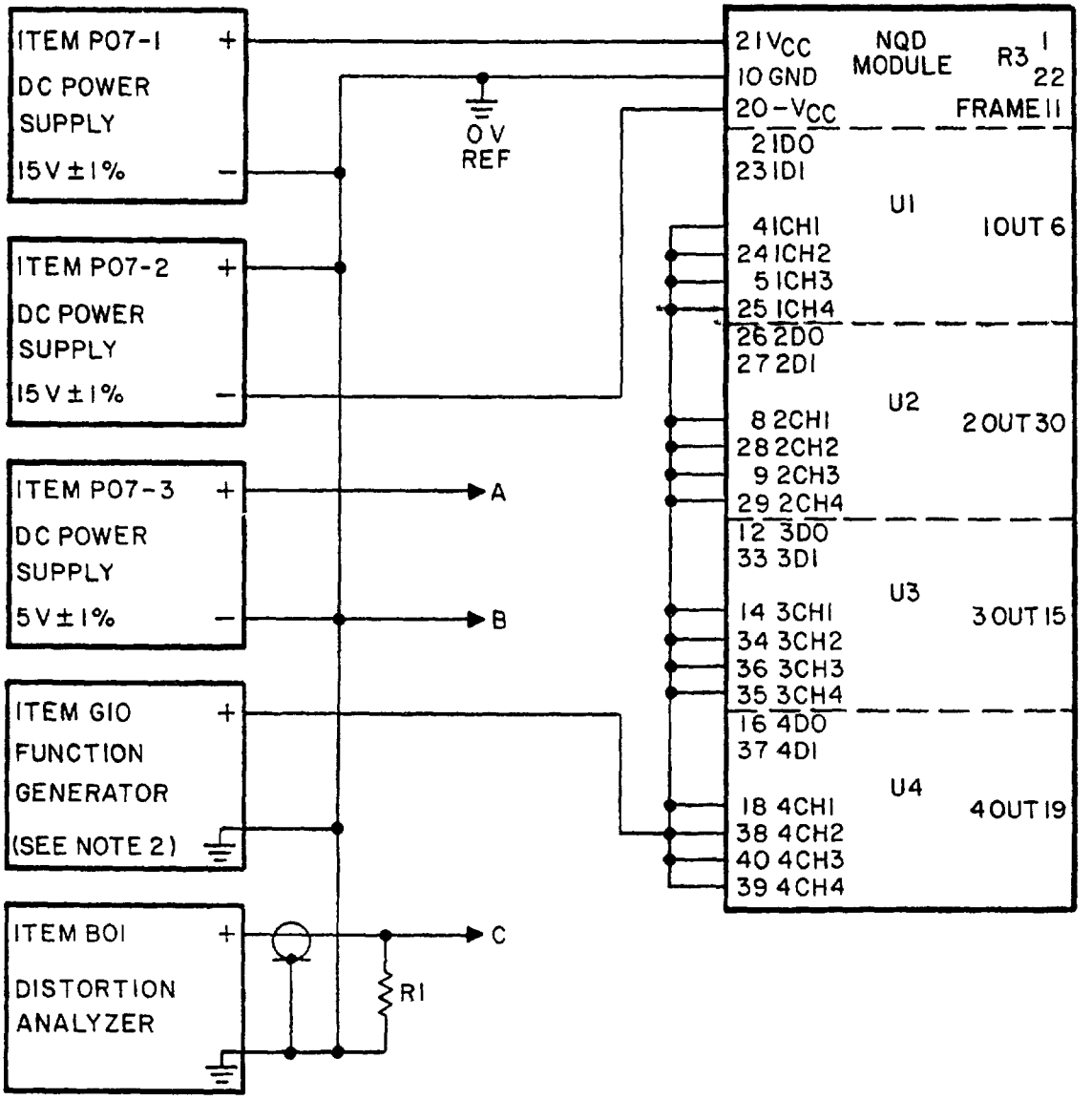
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TABLE XIII. Distortion test. 1/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Distortion	Open	2,23	6	0	1	%	15
M2	"	2	23	"	"	"	"	"
M3	"	23	2	"	"	"	"	"
M4	"	2,23	Open	"	"	"	"	"
M5	"	Open	27,26	30	"	"	"	"
M6	"	26	27	"	"	"	"	"
M7	"	27	26	"	"	"	"	"
M8	"	27,26	Open	"	"	"	"	"
M9	"	Open	33,12	15	"	"	"	"
M10	"	12	33	"	"	"	"	"
M11	"	33	12	"	"	"	"	"
M12	"	33,12	Open	"	"	"	"	"
M13	"	Open	16,37	19	"	"	"	"
M14	"	16	37	"	"	"	"	"
M15	"	37	16	"	"	"	"	"
M16	"	16,37	Open	"	"	"	"	"

1/ This test also verifies an output voltage swing of ±10 V.

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NOTES:
1. R1 = 2 kΩ ±5%, 1/4 W.
2. Item G10 setup:
   Frequency = 1 kHz ±10%.
   Amplitude = 7 V rms ±1%.
   Sine wave.

FIGURE 15. Distortion test circuit for table XIII.

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TABLE XIV. Crosstalk check test. 1/

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Crosstalk check	Open	2,23,4	24,5,25,8,28,9,29,14, 34,36,35,18,38,39,40	6	0	22	mV	16
N2	"	2	23,24	4,5,25,8,28,9,29,14, 34,36,35,18,38,39,40	"	"	"	"	"
N3	"	23	2,5	4,24,25,8,28,9,29,14, 34,36,35,18,38,39,40	"	"	"	"	"
N4	"	2,23	25	4,5,24,8,28,9,29,14, 34,36,35,18,38,39,40	"	"	"	"	"
N5	"	Open	26,27,8	4,24,5,25,9,29,14,34, 36,35,28,18,38,39,40	30	"	"	"	"
N6	"	26	27,28	4,24,5,25,8,9,29,14, 34,36,35,18,38,40,39	"	"	"	"	"
N7	"	27	26,9	4,24,5,25,8,28,29,14, 34,36,35,18,38,40,39	"	"	"	"	"
N8	"	27,26	29	4,24,5,25,8,9,28,14, 34,36,35,18,38,40,39	"	"	"	"	"
N9	"	Open	12,33,14	4,24,5,25,8,9,28,29, 34,36,35,18,38,40,39	15	"	"	"	"
N10	"	12	33,34	4,24,5,25,8,9,28,29, 14,36,35,18,38,40,39	"	"	"	"	"
N11	"	33	12,36	4,24,5,25,8,9,28,29, 14,34,35,18,38,40,39	"	"	"	"	"
N12	"	12,33	35	4,24,5,25,8,9,28,29, 14,34,36,18,38,40,39	"	"	"	"	"
N13	"	Open	16,37,18	4,24,5,25,8,9,28,29, 14,34,36,35,38,39,40	19	"	"	"	"

See footnote at end of table.

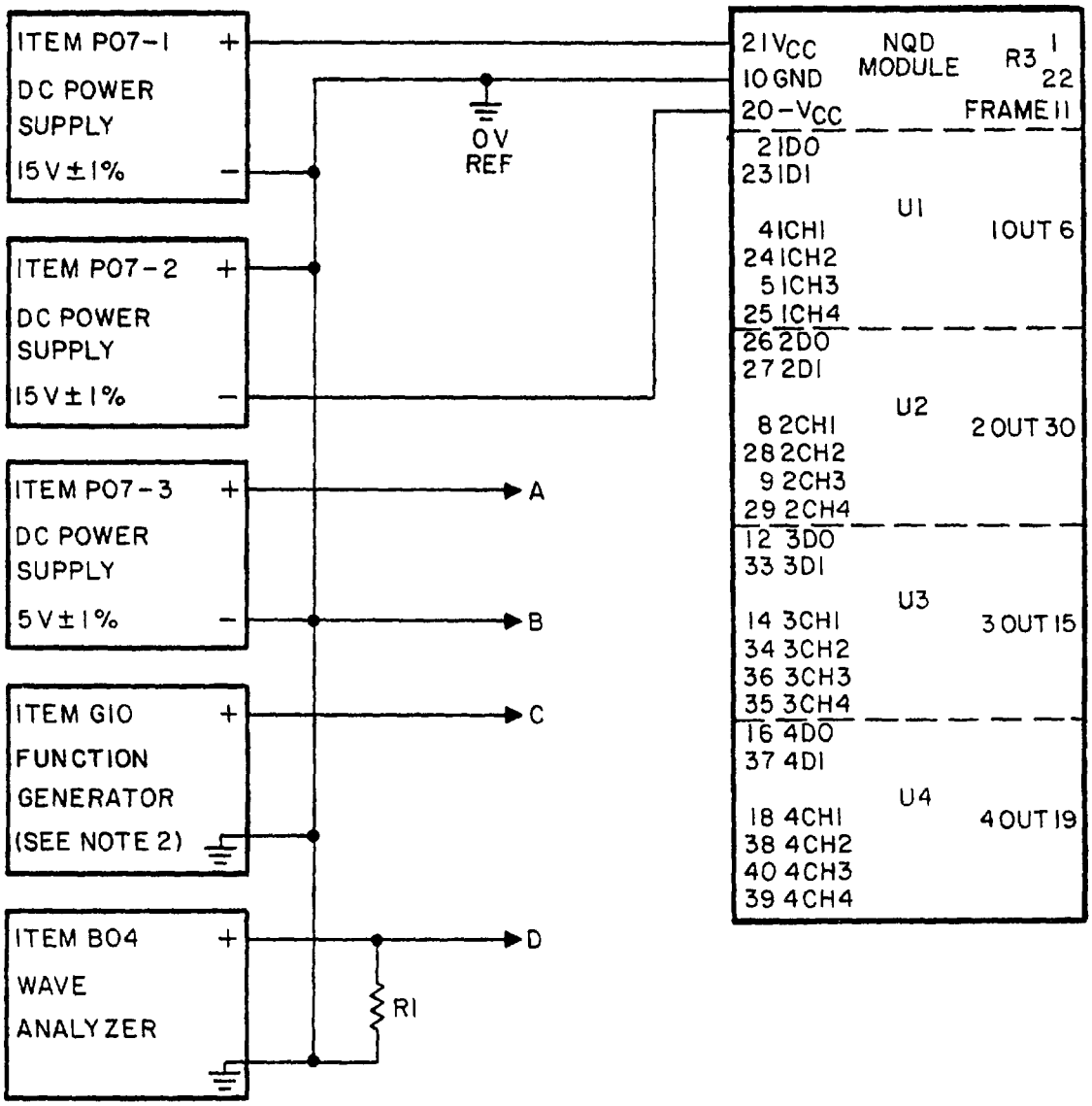
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TABLE XIV. Crosstalk check test 1/ - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N14	Crosstalk check	16	37,38	4,24,5,25,8,9,28,29, 14,34,36,18,35,39,40	19	0	22	mV	16
N15	"	37	16,40	4,24,5,25,8,9,28,29, 14,34,36,18,35,38,39	"	"	"	"	"
N16	"	16,37	39	4,24,5,25,8,9,28,29, 14,34,36,18,35,38,40	"	"	"	"	"

1/ Crosstalk (table I, dB) = $20 \log \left(\frac{\text{Item B04 Reading}}{7 \text{ V rms}} \right)$.

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NOTES:
1. R1 = 2 kΩ ±5%, 1/4 W.
2. Item G10 setup:
   Frequency = 5 kHz ±1%.
   Amplitude = 7 V rms ±1%.
   Sine wave.
3. Set item B04 bandwidth to 10 Hz. Tune to 5 kHz and fine tune for a maximum deflection.

FIGURE 16. Crosstalk check test circuit for table XIV.

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TABLE XV. Functional test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Functional	4	24,5,25	2,23	6	2.35	2.45	V	17
P2	"	2,24	4,5,25	23	"	"	"	"	"
P3	"	23,5	4,24,25	2	"	"	"	"	"
P4	"	2,23,25	4,24,5	Open	"	"	"	"	"
P5	"	8	28,9,29	26,27	30	"	"	"	"
P6	"	26,28	8,9,29	27	"	"	"	"	"
P7	"	27,9	8,28,29	26	"	"	"	"	"
P8	"	26,27,29	8,28,9	Open	"	"	"	"	"
P9	"	14	34,36,35	12,33	15	"	"	"	"
P10	"	12,34	14,36,35	33	"	"	"	"	"
P11	"	33,36	14,34,35	12	"	"	"	"	"
P12	"	12,33,35	14,34,36	Open	"	"	"	"	"
P13	"	18	38,40,39	16,37	19	"	"	"	"
P14	"	16,38	18,40,39	37	"	"	"	"	"
P15	"	37,40	18,38,39	16	"	"	"	"	"
P16	"	16,37,39	18,38,40	Open	"	"	"	"	"

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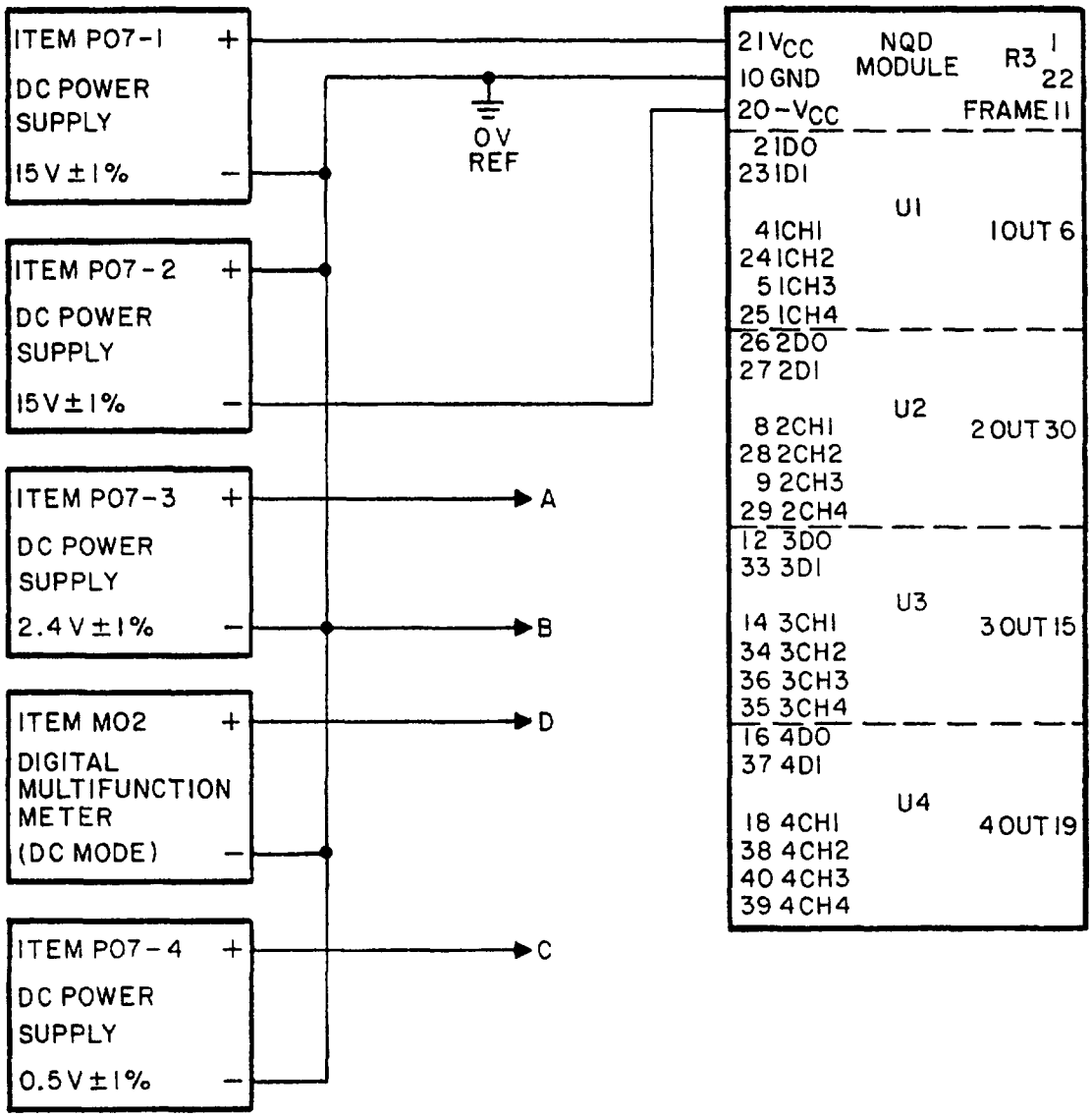


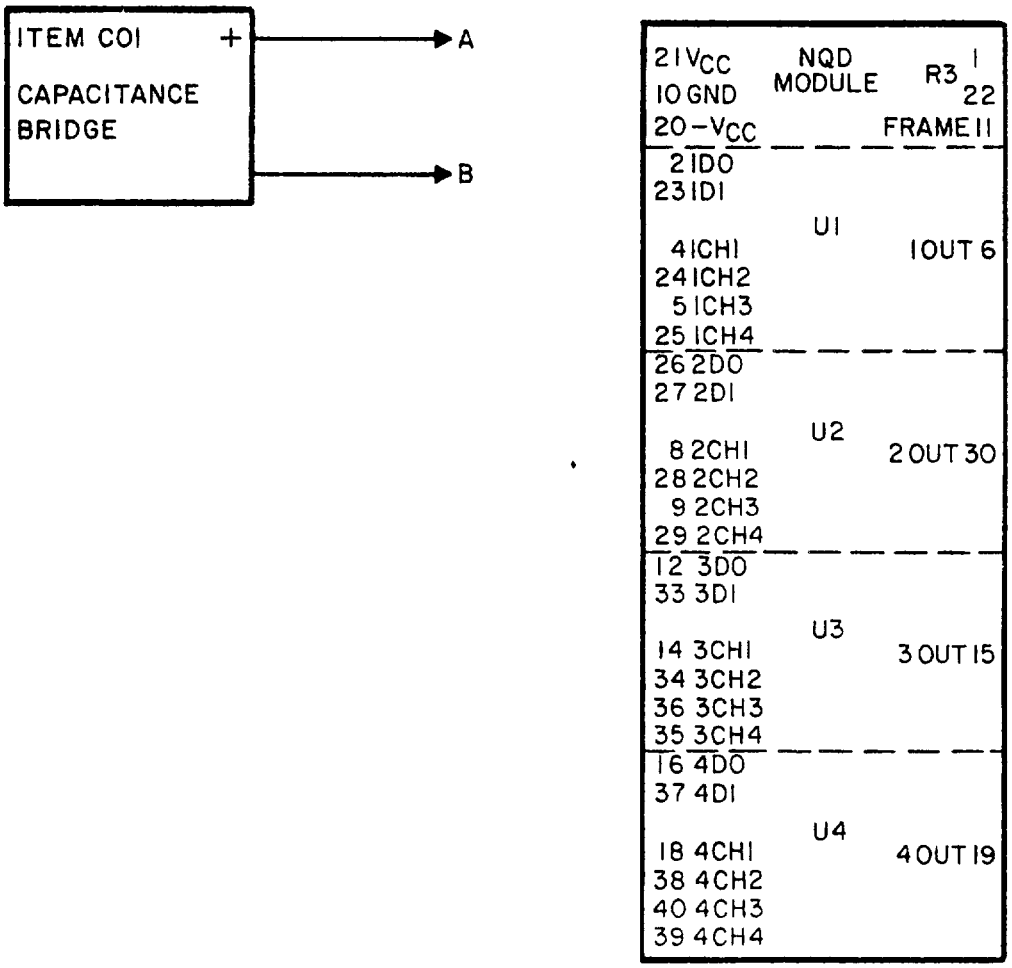
FIGURE 17. Functional test circuit for table XV.

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TABLE XVI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	Power supply filter capacitance	21	10	0.75	1.25	µF	18
Q2	Power supply filter capacitance	10	20	0.75	1.25	µF	18

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NOTE: Item C01 setup:
Frequency = 120 Hz ±10%.
Vpk = 0.3 V maximum.

FIGURE 18. Power supply filter capacitance test circuit for table XVI.

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